MIL-R-18265A(WEP)
20 FEBRUARY 1961
SUPERSEDING
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15 NOVEMBER 1954

MILITARY SPECIFICATION

RECEIVING SET, RADIO AN/ARR-41 (*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

*1.1 Scope - The equipment covered by this specification shall receive radio telephony (A-3), MCW (A-2), and CW telegraphy (A-1). It shall be suitable for installation in military aircraft and for local operation and shall require no remote control facilities.

*1.2 Classification - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Appl. Para.
Receiver, Radio	R-648/ARR-41	3.4.1
Mounting	MT-1518/ARR-41	3.4.1

*1.3 Associated Equipment - When installed in aircraft, the radio receiving set will be connected to and used with equipment generally similar to that listed below. These associated equipments are not a part of the radio receiving set and shall not be furnished by the contractor as part of the equipment.

Antenna - Fixed Aircraft
Headset - H-173/AR or equivalent
Interconnecting Box - J-1013/AIC
Intercommunication Set - AN/AIC-14 or equivalent
Antenna Coupler - CU-351/AR (Optional Use Only)

2. APPLICABLE DOCUMENTS

FSC 5821

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*2.1 General - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-17	Cases; Bases, Mounting; and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-17555	Preparation for Delivery of Electronic, Electrical, and Electro-Mechanical Equipment and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, or 115/200 Volt 3-Phase AC, or Combinations

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PUBLICATIONS

Bureau of Naval Weapons

SAR-300 Material Changes and Material Bulletins, Preparation of

*2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of Parts and Materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following factors shall govern.

(1) Parts and Materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the contractors can demonstrate that the use of standard parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specification.

(3) When a non-standard part is used, the meeting of general material requirements such as non-flammability, fungus resistance, non-toxicity, etc., and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed so that the unit as a whole will pass material, electrical and environmental tests, without consideration to the quality of the detail parts. Any assembly built to be expended shall be first approved by the procuring activity.

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*3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

*3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.1.3 Controls

3.1.3.1 Tuning Control - The tuning control need not rotate only in a clockwise direction for an increasing frequency characteristic as required by Specification MIL-E-5400.

3.1.3.2 Tuning Dials - When a counter type dial, indicating operating frequency to the nearest kilocycle, is used, visibility of two calibration numbers at any dial setting per Specification MIL-E-5400 is not required.

3.1.4 Finishes

3.1.4.1 Brass - Brass hardware and brass gears used as bearing or sliding surfaces shall not have a protective finish as required by MIL-E-5400.

3.1.4.2 External Fasteners - External front panel fasteners used to secure the front panel to the chassis shall be labelled to indicate the direction of rotation for locking and the color contrast requirements of Specification MIL-E-5400 shall not apply.

3.1.5 Inductors

3.1.5.1 Stability of Q - Every precaution shall be taken to maintain stability of Q in radio frequency inductors to insure proper performance of equipment. The stability of Q requirements specified in Specification MIL-E-5400 shall not apply provided the equipment satisfies all of the requirements for operation under Service Conditions as specified herein.

3.1.6 Marking and Identification

3.1.6.1 Panel Markings - Markings which will permit the operator to set the r-f and audio gain control to a predetermined setting in accordance with Specification MIL-E-5400 is not required. Likewise, the BFO control shall have only a mark to identify the center of its range.

3.1.6.2 Receptacle Identification - Identification of receptacles is required where space is available. In the case of plug-in subassemblies the visibility requirement is applicable only when the subassembly is removed. Such identification is always required on at least one side of the panel on which such receptacles are mounted.

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3.1.6.3 Tube Socket Identification - Tube socket identification shall be in accordance with Specification MIL-E-5400 except where space is limited or not available. Such identification is always required on at least one side of the panel on which such sockets are mounted.

3.1.7 Rotary Switch Mounting - In addition to the requirements of Specification MIL-E-5400, rotary switches may be braced to meet the vibration and shock requirements of this specification. Bracing shall be effected by attachment of braces to the through bolts at the rear of the switch.

*3.2 General Requirements - This equipment shall fulfill all applicable design, construction and workmanship requirements of Specification MIL-E-5400 unless otherwise specified herein.

*3.2.1 Standard Conditions - The following conditions shall be used to establish normal performance characteristics and for making laboratory bench tests on the equipment:

Temperature	Room ambient ($30^{\circ} \pm 10^{\circ}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ± 0.5 V DC

*3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combination of service conditions as specified in Specification MIL-E-5400 for Class 1 equipment.

*3.2.2.1 Vibration - The equipment shall not be required to meet the provisions of 5-500 cps vibration with vibration isolators removed as specified in Specification MIL-E-5400 Curve II.

3.2.3 Total Weight - The total weight of the equipment excluding cables shall not exceed 40 pounds.

3.2.4 Operational Stability - The equipment shall operate with optimum performance for 100 hours, continuously or intermittently, without necessity for readjustment of any controls which are non-accessible during flight.

3.2.5 Operating Life

3.2.5.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

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3.2.5.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.6 Interconnecting Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

*3.2.7 Cables and Connectors - The equipment shall provide for the use of cables and external connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and that portion of connectors which is to be attached to the cables shall not be supplied as part of the equipment.

*3.2.8 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power source with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts. (See 6.5)

DC Power 28V, Category "B," 5.8 Amps

*3.2.9 Input Power Connections - Input power connections shall be in accordance with MS 25245

3.2.10 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.10.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

*3.2.11 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

*3.2.12 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

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*3.2.13 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.14 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.15 Warm-Up Time - The equipment shall be allowed a 20 minute warm-up period in the "Voice" position before conducting any of the tests as specified in Specification MIL-T-5422.

3.2.16 Test Points - Double height lugs suitably labelled shall be provided as test points on terminal boards in lieu of the chassis test points requirement of Specification MIL-E-5400.

*3.3 Performance

*3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Frequency Coverage - By means of the front panel controls, it shall be possible to select any frequency within the following ranges:

190 to 550 KC

2.0 to 25.0 MC

Voice, CW or MCW reception shall be available on all frequencies.

3.3.3 Sensitivity, Phone - The sensitivity of the receiver over the specified frequency range shall be such that an r-f input of 5 microvolts modulated 30% at 1000 cps produces an audio output into the specified load of at least 100 milliwatts with a signal plus noise-to-noise ratio of at least 6 db. The sensitivity under service conditions shall not vary by more than 10 db from that specified above. The audio power output of the receiver shall be adjusted to 500 milliwatts (-10%, +20%) with an r-f input of 1000 microvolts 30% modulated at 1000 cps under standard test conditions.

3.3.4 Sensitivity, CW - The sensitivity of the receiver over the specified frequency range shall be such that an r-f input of 5 microvolts, with the beat frequency oscillator control adjusted for 1000 cps audio output, shall produce an audio output of at least 100 mw into the specified load with a signal plus noise-to-noise ratio of at least 10 db. The sensitivity under service conditions shall not vary by more than 10 db from that specified above.

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*3.3.5 Selectivity - The receiver shall be operable in either of two conditions of selectivity as set forth below:

Attenuation Below Resonant Freq.	Normal Selectivity Bandwidth	Sharp Selectivity Bandwidth
6 db	5.0 kc min.	1.2 kc min.
60 db	20.0 kc max.	5.0 kc max.

The change in selectivity required by the two operating conditions specified above shall be effected in the i-f amplifier prior to the audio detector. It shall be possible to select either of the two conditions for all required types of reception.

3.3.6 Image and Other Spurious Signal Rejection - The receiver shall provide an attenuation of not less than 60 db, relative to a 5 microvolt r-f signal input at the resonant frequency, to all image frequencies and to all other spurious responses including those resulting from the combination of fundamental and harmonic frequencies of the conversion oscillator with the undesired signals.

3.3.7 I-F Rejection - The receiver shall provide an attenuation of not less than 80 db, relative to a 5 microvolt r-f input signal at the resonant frequency, to signals on the frequencies of the intermediate frequency amplifiers.

*3.3.8 AVC Characteristics - Under standard conditions, the automatic volume control action shall be such that when a 1,000 microvolt signal, modulated 30% at 1000 cps, is applied, the audio output power shall be 500 milliwatts (-10%, +20%). When the modulated signal is varied from 10 to 100,000 microvolts, the audio power output shall not vary more than ±2 db from the 500 milliwatt reference level on all bands with the exception of the 2.0 to 4.0 mc range. On this band the audio output power may vary ± 3 db from the 500 milliwatt reference level. The audio output power of the receiver shall not change more than 6 db with input signals up to 2 volts. Under service conditions, the 500 mw reference level obtained with the 1000 microvolt signal modulated 30% at 1000 cps shall not vary more than minus 6 db nor more than plus 3 db. Under service conditions, the specified regulation of audio output shall be obtained with input variations of 30 to 100,000 microvolts.

3.3.9 Regeneration - The interstage and overall regeneration of the receiver shall be so minimized that no evidence of oscillation, instability, or excessive regeneration shall occur at maximum gain control settings under either standard or service conditions when the DC input voltage is raised to 30 volts.

*3.3.10 Noise Limiting - The second detector of the receiver (that which follows the fixed frequency intermediate frequency amplifier) shall be provided with an instantaneous peak limiter. The noise limiting action shall effectively limit or reduce voltage surges or peaks exceeding the receiver output obtained with 40 to 70% modulated signal at maximum sensitivity. Its action shall be relatively instantaneous so as not to obliterate signals by a permanent reduction in receiver gain under the influence of continuous r-f noises such as are caused by commutation, ignition systems, etc.

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3.3.11 Audio Characteristics

3.3.11.1 Capability - Under standard conditions, the receiver shall be capable of 500 milliwatts (-10%, +20%) for an r-f input of 1000 microvolts modulated 30% at 1000 cps. The total harmonic content of the receiver audio output shall not exceed 10% at the 500 milliwatt level with an r-f input having any level between 100 and 100,000 microvolts and modulated 30% at 1000 cps. Under service conditions, the output shall not be more than 6 db below that obtained under standard conditions nor more than 3 db above.

*3.3.11.2 Overall Fidelity - The overall audio response shall not vary more than plus or minus 3 db relative to the response at 1000 cps for modulation frequencies of 300 to 2200 cps. Frequencies below 300 and above 3000 cps shall be attenuated the maximum practicable amount without recourse to inclusion in the receiver of large and/or heavy filter components. Distortion measurements are to be made with modulation frequencies in the range of 300 to 2200 cps only.

3.3.12 CW Overload Characteristics - With the receiver adjusted for CW reception (the bfo control adjusted for an audio output of approximately 1000 cps) at an r-f input of any level between 10 and 1000 microvolts, the r-f gain control shall be adjusted to give an audio output of 500 milliwatts. A further increase in the r-f input level of 40 db shall not cause a change in the receiver output level in excess of ±6 db.

3.3.13 Gain Control Characteristics

3.3.13.1 R-F Gain Control - The r-f gain control shall be ganged with the audio gain control and shall be effective in calibrate position and on CW and MCW reception. It shall provide an approximately linear variation of not less than 40 db in the overall gain of the receiver from its r-f input to its a-f output. The sensitivity control, effective on voice reception, shall have such a range as to permit a variation in the sensitivity of the receiver from the maximum to a value between 20 and 100 microvolts.

3.3.13.2 Audio Gain Control - The audio gain control shall be ganged to the r-f gain control and shall permit a variation in audio output from the maximum to a value at least 40 db below maximum. The audio gain control shall be effective on voice reception only.

*3.3.14 CW Characteristics - A control shall be provided on the front panel of the receiver for varying the CW beat frequency over a range of not less than ±2000 cps nor more than ±4000 cps under standard conditions and not less than ±2000 cps nor more than ±5000 cps under service conditions.

3.3.15 Frequency Stability - After 20 minutes of warmup and over a full range of standard or service conditions, the frequency excursion (including initial calibration error) from the selected frequency shall not exceed the following:

190 - 550 kc - ±2.1 kc

2.0 - 25.0 mc - ±(1.9 kc + .01%)

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*3.3.16 FSK Requirements - The receiver, when suitably interconnected with a separate keying unit, shall operate a standard teletype printer when receiving a standard FSK signal (850 cps shift) at keying rates up to 75 cps. When the receiver is so operated, the input to auxiliary teletype equipment shall be in the form of two-tone audio signals. Testing of the requirements of this paragraph shall only be performed when using suitable government furnished keying adapter, teleprinter and signal source equipment.

3.3.17 Tuning and Adjustment - With the equipment operating properly, no test equipment, tools, or other means or devices external to the equipment shall be required for its proper tuning and adjustment to any assigned frequency within the specified frequency range.

3.4 Detail Requirements

3.4.1 Receiver, Radio R-648/ARR-41

3.4.1.1 Function - The radio receiver shall receive r-f energy from the antenna, amplify it, and convert it to aural intelligence.

*3.4.1.2 Form Factor and Mounting - The size of the receiver case, including louvers and front panel handles shall not exceed 7-11/16 by 17-1/8 by 13-3/8 inches. The size of Mounting MT-1518/ARR-41 in its unloaded condition, shall not exceed 1-15/16 by 17 by 13-3/8 inches. The overall dimensions when the radio receiver is attached to the mounting base and mounted on a horizontal plane surface shall not exceed 9-1/2 by 17-1/8 by 13-1/2 inches.

3.4.1.3 Contents - This unit shall contain the complete receiver including the frequency generating and selecting mechanisms, power supply, and all other parts necessary to make the equipment function as a unit. In addition, the receiver shall be contained in a single case capable of being mounted in an aircraft by means of Mounting MT-1518/ARR-41. The mounting base shall be so arranged as to furnish all support required by the receiver. The requirements of Specification MIL-C-172 shall apply except for the dimensional requirements as specified herein. It shall be possible to remove the receiver from the mounting rack and to remove the dust cover using only ordinary hand tools without requiring removal of any cable other than the front panel connectors. It shall be possible to remove or replace any plug-in-subassembly located on the main chassis using only ordinary hand tools, and without unsoldering any connections. It shall also be possible to replace any plug-in-subassembly with another of the same type without resynchronization except that realignment of the tuning elements may be necessary. All filter components used to reduce conducted and radiated noise on the power cables shall be integral with the receiver.

3.4.1.4 Weight With Mounting - The combined weight of the receiver unit and its mounting base shall not exceed 40.0 pounds.

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3.4.1.5 Electrical Connections - The following connection facilities shall be provided on the front panel of the receiver:

(1) A coaxial connector, type BNC, for connection of the receiver input circuits to an antenna.

(2) A type JK-34 jack, or equivalent, for headset connection.

(3) A type AN plug for input power and audio output connections.

3.4.1.6 Controls, Internal - All semi-fixed adjustable devices provided for tracking, alignment, gain, etc. shall be readily accessible when the unit is removed from its case. If tools other than a conventional straight-lipped screwdriver are required for manipulation of these devices, such tools shall be furnished and clipped to the unit inside the case.

3.4.1.7 Controls, Panel Mounted - The following controls shall be located and suitably marked on the front panel:

(1) A rotary control for selection of the proper frequency band.

(2) A rotary control for the fine adjustment of frequency. Dials associated with these tuning controls shall indicate the frequency at least within the nearest kilocycle.

(3) A rotary control to adjust the audio output during "voice" reception and the sensitivity of the receiver during CW and MCW reception.

(4) A rotary control for varying the pitch of the beat note when receiving CW signals. This control shall be effective during CW and calibration operation only.

(5) A two-position rotary switch for control of primary to the receiver.

(6) A five-position rotary switch for selection of Calibrate (CAL), VOICE, VOICE SHARP, CW, CW SHARP(CWS). MCW operation shall be available on the Voice positions.

(7) A screwdriver adjustable rotary control for adjusting the threshold of sensitivity of the receiver.

3.4.1.8 Terminations

3.4.1.8.1 R-F Termination - The r-f input to the receiver shall be through a 50-ohm (resistive only) dummy antenna. This value of 50 ohms shall include the internal impedance of the signal generator.

3.4.1.8.2 Audio Terminations - The audio output (headset) of the receiver shall be terminated in a 300 ohm non-inductive load which shall include means for measuring the audio frequency power output.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.7. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction

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model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

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(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1

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Quantity of Equipments Offered for Acceptance	Quantity to be Tested
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

(3) Manufacturing reliability test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not

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applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following: When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)

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Humidity	Room ambient
A.C. voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stablize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

*4.6 Reconditioning of Acceptance Test Samples - Each equipment which has been subjected to acceptance tests (Sampling, special and life tests) shall be reconditioned as follows:

(1) On completion of the Sampling, Special or Life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

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4.7 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.10 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

*5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

*6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

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(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.13. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 The primary input power requirements defined in 3.2.8 changes the requirements as defined in Specification MIL-E-7894 to requirements as defined in MIL-STD-704. Properly assigned category will require no design changes of the equipment.

6.6 An asterisk (*) preceding a paragraph number denotes a change in that paragraph from the original specification.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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